Google Cloud H2S

Gen AI Academy

APAC Edition

Build in APAC. Build for the world.

Participant Details

a. Participant Name: Mrithula KL
b. Problem Statement: Hackathon - SecondBrain

Brief about the idea

Modern professionals manage tasks, schedules, notes and research across multiple disconnected tools. This leads to context switching, missed deadlines and fragmented knowledge.

Second Brain is a **multi-agent AI productivity system** that acts as an intelligent assistant capable of:

- Managing tasks automatically
- Scheduling events intelligently
- Storing structured notes
- Performing contextual research
- Maintaining memory of user workflows

Built using:
Agent Development Kit (ADK)
Model Context Protocol (MCP)
AlloyDB AI

It creates a **centralized intelligent productivity layer** instead of another standalone app.

From Problem Statement to Working Solution:

Problem Statement
Users today rely on multiple tools:
Task managers
Calendars
Notes apps
Search engines

But they **don't communicate with each other intelligently**.
Result:
❌ fragmented workflow
❌ missed priorities
❌ manual scheduling
❌ no contextual memory

From Problem Statement to Working Solution:

I've built a **multi-agent orchestration system** where:
Orchestrator Agent understands intent; Specialized agents execute actions
MCP enables tool integration; AlloyDB stores structured memory
Instead of one assistant → we use **collaborative agents**

Real-World Impact
Second Brain helps users:
✅ plan their week automatically
✅ detect scheduling conflicts
✅ retrieve contextual notes instantly
✅ maintain long-term productivity memory
✅ reduce decision fatigue

Especially useful for students, developers, researchers , working professionals

From Problem Statement to Working Solution:

Core Workflow

User prompt → Orchestrator Agent
↓
Intent detection
↓
Task / Calendar / Notes / Research Agent activation
↓
Execution via MCP tools
↓
Stored inside AlloyDB structured memory
↓
Context-aware response returned

Opportunities

- How different is it from any of the other existing ideas?
- USP of the proposed solution

Second Brain is:
🧠 agent-driven instead of UI-driven
🔗 cross-tool aware
📊 structured-memory powered
⚡ intent-aware
📅 conflict-detecting
📚 research-integrated

Most assistants: react
Second Brain: **plans proactively**

List of features offered by the solution

Multi-Agent Orchestration: Routes requests intelligently to the correct agent

Task Intelligence: Create and update tasks, track deadlines and detect overdue items

Smart Calendar Scheduling: Auto-create events, detect conflict, suggest free time slots

Contextual Notes System: Store and retrieve notes intelligently, attach notes to events/tasks

Research Assistant: Web search integration, summarization, context-aware responses

Persistent Agent Memory: Stores preferences, context, previous plans, productivity history inside AlloyDB.

Process flow diagram or Use-case diagram

- User Input
- Orchestrator Agent
- Intent Classification
- Specialised Agent Selection
- Execution via MCP Tools
- Store in AlloyDB
- Return smart response

Architecture diagram of the proposed solution:

```
Frontend / API Layer
      ↓
Orchestrator Agent
      ↓
--------------------------
| Task Agent             |
| Calendar Agent         |
| Notes Agent            |
| Research Agent         |
--------------------------
      ↓
MCP Tool Layer
      ↓
AlloyDB Database
```

Technologies/Google Services used in the solution

Core AI Framework
Agent Development Kit (ADK)
Model Context Protocol (MCP)
AlloyDB AI

Backend
FastAPI

Frontend
HTML + JS lightweight interface

Cloud Platform
Google Cloud

Case I:

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QUICK PROMPTS

- I have a project deadline Friday — set up my week
- Schedule a team sync tomorrow at 3pm and add prep notes
- What tasks are overdue?
- Create a 30-day learning plan for system design
- Find free time for a 2-hour deep work block this week

SYSTEM

Active tasks
84
in AlloyDB

Events this week
60
upcoming

Notes
50+
saved

AGENTS

`task_agent` `calendar_agent`

SB Second Brain

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- **Set up your week** around a deadline
- **Schedule events** and block focus time
- **Save notes** from a meeting or research session
- **Build a learning plan** on any topic

What would you like to accomplish?

You You

What tasks are overdue and what upcoming events should i prioritize this week?

SB Second Brain

Here's an overview of your overdue tasks. I couldn't retrieve prioritized upcoming events for this week, as I need specific start and end dates to find them.

Summary
You have several overdue tasks across various projects that require attention.

Tell me what you need to accomplish...

Enter to send · Shift+Enter for new line

Tasks | Events | Notes

- Weekly plan — 8 Apr, 14:45
- Weekly plan — 8 Apr, 14:45
- Weekly plan — 8 Apr, 14:45
- Meeting notes: Team standup — 8 Apr, 14:45
- Weekly plan — 8 Apr, 14:45
- Meeting notes: Deep work block — 8 Apr, 14:45
- Weekly plan — 8 Apr, 14:45
- Reference: onboarding setup — 8 Apr, 14:45
- Research: API — 8 Apr, 14:45
- Weekly plan — 8 Apr, 14:45
- Reference: caching setup — 8 Apr, 14:45

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Overdue Tasks

* **Design notifications architecture** (Personal) - Due: April 1st
* **Schedule caching sync with team** (Q2 Marketing) - Due: April 2nd
* **Research notifications alternatives** (Team OKRs) - Due: April 2nd
* **Schedule CI/CD sync with team** (Website Redesign) - Due: April 2nd
* **Write caching documentation** (Personal) - Due: April 3rd
* **Research reporting alternatives** (Website Redesign) - Due: April 3rd
* **Write caching documentation** (Q2 Marketing) - Due: April 3rd
* **Design caching architecture** (API Migration) - Due: April 3rd
* **Refactor onboarding module** (Learning) - Due: April 3rd
* **Research billing alternatives** (Learning) - Due: April 4th
* **Set up onboarding monitoring** (Team OKRs) - Due: April 4th
* **Set up billing monitoring** (Website Redesign) - Due: April 4th
* **Test CI/CD edge cases** (Q2 Marketing) - Due: April 5th
* **Update analytics dependencies** (Q2 Marketing) - Due: April 5th
* **Write onboarding documentation** (Team OKRs) - Due: April 5th
* **Review billing pull request** (Learning) - Due: April 5th
* **Test analytics edge cases** (Learning) - Due: April 5th
* **Prepare reporting presentation** (Q2 Marketing) - Due: April 6th
* **Write API documentation** (Team OKRs) - Due: April 7th
* **Write search documentation** (Team OKRs) - Due: April 7th
* **Refactor notifications module** (API Migration) - Due: April 7th
* **Prepare notifications presentation** (Personal) - Due: April 8th

Tell me what you need to accomplish...

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Tasks | Events | Notes

- Weekly plan — 8 Apr, 14:45
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- Weekly plan — 8 Apr, 14:45
- Meeting notes: Team standup — 8 Apr, 14:45
- Weekly plan — 8 Apr, 14:45
- Meeting notes: Deep work block — 8 Apr, 14:45
- Weekly plan — 8 Apr, 14:45
- Reference: onboarding setup — 8 Apr, 14:45
- Research: API — 8 Apr, 14:45
- Weekly plan — 8 Apr, 14:45
- Reference: caching setup — 8 Apr, 14:45
- Research: CI/CD

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* **Refactor onboarding module** (Learning) – Due: April 3rd
* **Research billing alternatives** (Learning) – Due: April 4th
* **Set up onboarding monitoring** (Team OKRs) – Due: April 4th
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* **Refactor notifications module** (API Migration) – Due: April 7th
* **Prepare notifications presentation** (Personal) – Due: April 8th
* **Write authentication documentation** (Team OKRs) – Due: April 8th

Please let me know if you'd like to get a list of events for a specific date range!

▼ ▸ `orchestrator trace · 2 agents called`

1. Retrieve all overdue tasks.
2. Retrieve upcoming events for the current week to help prioritize.

`task_agent` `calendar_agent`

Tell me what you need to accomplish...

Enter to send · Shift+Enter for new line

Tasks | Events | Notes

- **Weekly plan** 8 Apr, 14:45
- **Weekly plan** 8 Apr, 14:45
- **Weekly plan** 8 Apr, 14:45
- **Meeting notes: Team standup** 8 Apr, 14:45
- **Weekly plan** 8 Apr, 14:45
- **Meeting notes: Deep work block** 8 Apr, 14:45
- **Weekly plan** 8 Apr, 14:45
- **Reference: onboarding setup** 8 Apr, 14:45
- **Research: API** 8 Apr, 14:45
- **Weekly plan** 8 Apr, 14:45
- **Reference: caching setup** 8 Apr, 14:45
- **Research: CI/CD**

Case II:

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QUICK PROMPTS

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task_agent | calendar_agent | notes_agent | research_agent

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- **Save notes** from a meeting or research session
- **Build a learning plan** on any topic

What would you like to accomplish?

You You

I have a project deadline on April 12th 2026 at 5pm. Based on my overdue tasks and my scheduled events this week, suggest what i should focus on first to prepare

SB Second Brain

Here's what you should focus on first to prepare for your project deadline on April 12th, 2026, 5 PM, based on your overdue tasks:

You have a significant number of overdue tasks, so prioritizing is key.

Tell me what you need to accomplish...

Enter to send · Shift+Enter for new line

Tasks | Events | Notes

- Design notifications architecture — high | in_progress | 1 Apr, 14:44
- Prepare frontend presentation — urgent | done | 2 Apr, 14:44
- Schedule CI/CD sync with team — medium | pending | 2 Apr, 14:44
- Schedule caching sync with team — medium | in_progress | 2 Apr, 14:44
- Update analytics dependencies — low | cancelled | 2 Apr, 14:44
- Research notifications alternatives — low | in_progress | 2 Apr, 14:44
- Refactor onboarding module — medium | in_progress | 3 Apr, 14:44
- Review reporting pull request — medium | done | 3 Apr, 14:44
- Research reporting alternatives — low | pending | 3 Apr, 14:44
- Write caching documentation — high | pending | 3 Apr, 14:44
- Design caching architecture — high | in_progress | 3 Apr, 14:44

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Prepare reporting presentation` (Due: 2026-04-06, Project: Q2 Marketing, Priority: Urgent)

* `Prepare notifications presentation` (Due: 2026-04-08, Project: Personal, Priority: Urgent)

* **Tackle High-Priority Overdue Tasks that Might Be Foundational:**
  * `Design notifications architecture` (Due: 2026-04-01, Project: Personal, Priority: High)
  * `Write caching documentation` (Due: 2026-04-03, Project: Q2 Marketing, Priority: High)
  * `Design caching architecture` (Due: 2026-04-03, Project: API Migration, Priority: High)
  * `Set up onboarding monitoring` (Due: 2026-04-04, Project: Team OKRs, Priority: High)
  * `Set up billing monitoring` (Due: 2026-04-04, Project: Website Redesign, Priority: High)
  * `Write onboarding documentation` (Due: 2026-04-05, Project: Team OKRs, Priority: High)
  * `Write API documentation` (Due: 2026-04-07, Project: Team OKRs, Priority: High)
  * `Write search documentation` (Due: 2026-04-07, Project: Team OKRs, Priority: High)
  * `Write authentication documentation` (Due: 2026-04-08, Project: Team OKRs, Priority: High)

To provide more tailored advice, please let me know the start and end dates for your current week so I can incorporate your scheduled events.

▼ ▸ orchestrator trace · 2 agents called

1. Retrieve all overdue tasks from the task agent.
2. Create a new task for the project deadline in the task agent.
3. Retrieve all scheduled events for the current week from the calendar agent.
4. Synthesize the overdue tasks, the new deadline, and weekly schedule to suggest initial focus areas for project preparation.

`task_agent` `calendar_agent`

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Case III:

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What would you like to accomplish?

You You

Create a 5-day learning plan for system design

SB Second Brain

Here is your 5-day learning plan for system design, which has also been saved for your reference!

This plan is designed to help you build a solid foundation in system design principles, common components, and practical application, progressing in complexity each day.

Tell me what you need to accomplish...

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5-Day System Design Learning Plan

Goal: Understand the fundamental principles, common components, and design considerations for building scalable, reliable, and maintainable software systems.

Day 1: Fundamentals & Core Principles

* **Key Topics:** Introduction to System Design, Core Pillars (Scalability, Reliability, Availability, Performance, Maintainability), Trade-offs, CAP Theorem, ACID vs. BASE, Latency vs. Throughput.
* **Key Concepts:** Monolith vs. Microservices, client-server model, stateless vs. stateful services.

Day 2: Data Storage & Management

* **Key Topics:** Database Choices (SQL vs. NoSQL use cases), Data Partitioning/Sharding, Replication & Consistency Models, Indexing, Caching (levels, strategies, invalidation).
* **Key Concepts:** Schema design, denormalization, read/write intensive systems, data locality.

Day 3: Core Components for Scalability & Communication

* **Key Topics:** Load Balancing, API Design (REST, RPC, GraphQL intro), Message Queues & Event Streams, Service Discovery, Content Delivery Networks (CDNs).
* **Key Concepts:** Horizontal scaling, decoupling, asynchronous communication, distributed transactions (briefly).

Day 4: Advanced Concepts & Operational Aspects

* **Key Topics:** Distributed Systems Challenges (fault tolerance, network partitions), Monitoring & Logging,

Tell me what you need to accomplish...

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Day 4: Advanced Concepts & Operational Aspects

* **Key Topics:** Distributed Systems Challenges (fault tolerance, network partitions), Monitoring & Logging, Security Considerations (Auth, AuthZ, Encryption), Deployment Strategies (Blue-Green, Canary), Cost Optimization.
* **Key Concepts:** Observability, reliability engineering, defense in depth.

Day 5: Putting It All Together & Practice

* **Key Topics:** System Design Interview Approach, Designing Sample Systems (e.g., URL Shortener, Twitter Timeline), Diagramming Tools & Techniques, Refinement & Iteration.
* **Key Concepts:** Iterative design, back-of-the-envelope estimations, communication of design choices.

General Recommended Resources (Throughout the Plan):

* **Books:** "Designing Data-Intensive Applications" by Martin Kleppmann, "System Design Interview" by Alex Xu.
* **Online Courses:** Platforms like Educative.io, Udemy, Coursera.
* **Blogs/Articles:** Engineering blogs of major tech companies (Netflix, Google, Meta, Amazon), Medium articles on specific topics.
* **YouTube Channels:** ByteByteGo, Gaurav Sen, Hussein Nasser.

▸ ▸ orchestrator trace · 2 agents called

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Future Enhancements

- Voice interaction
- Google Calendar live sync
- Email summarization agent
- Personal productivity analytics
- Reminder prediction engine
- Mobile app version

Public Cloud Run: https://secondbrain-1024066444984.us-central1.run.app
Repository: https://github.com/mrithu/SecondBrain
Demo: https://drive.google.com/file/d/10Wp3IANVp--_2_0sOgR2alaaCcedDFQw/view?usp=sharing

Google Cloud H2S

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Thank you